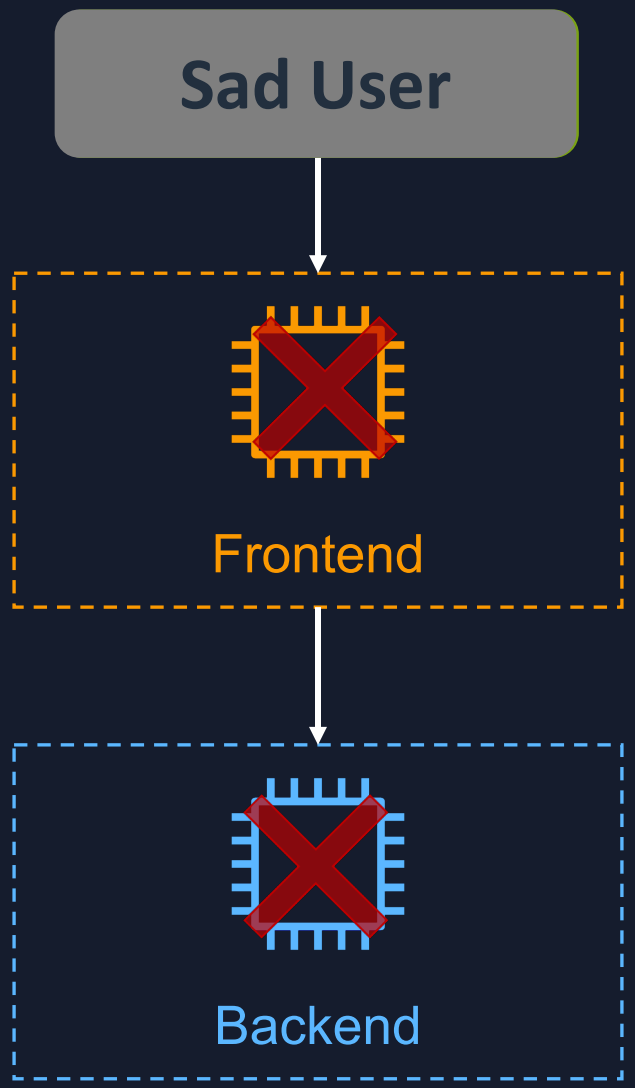
Amazon Web Services: Decoupling Workflows

Topics

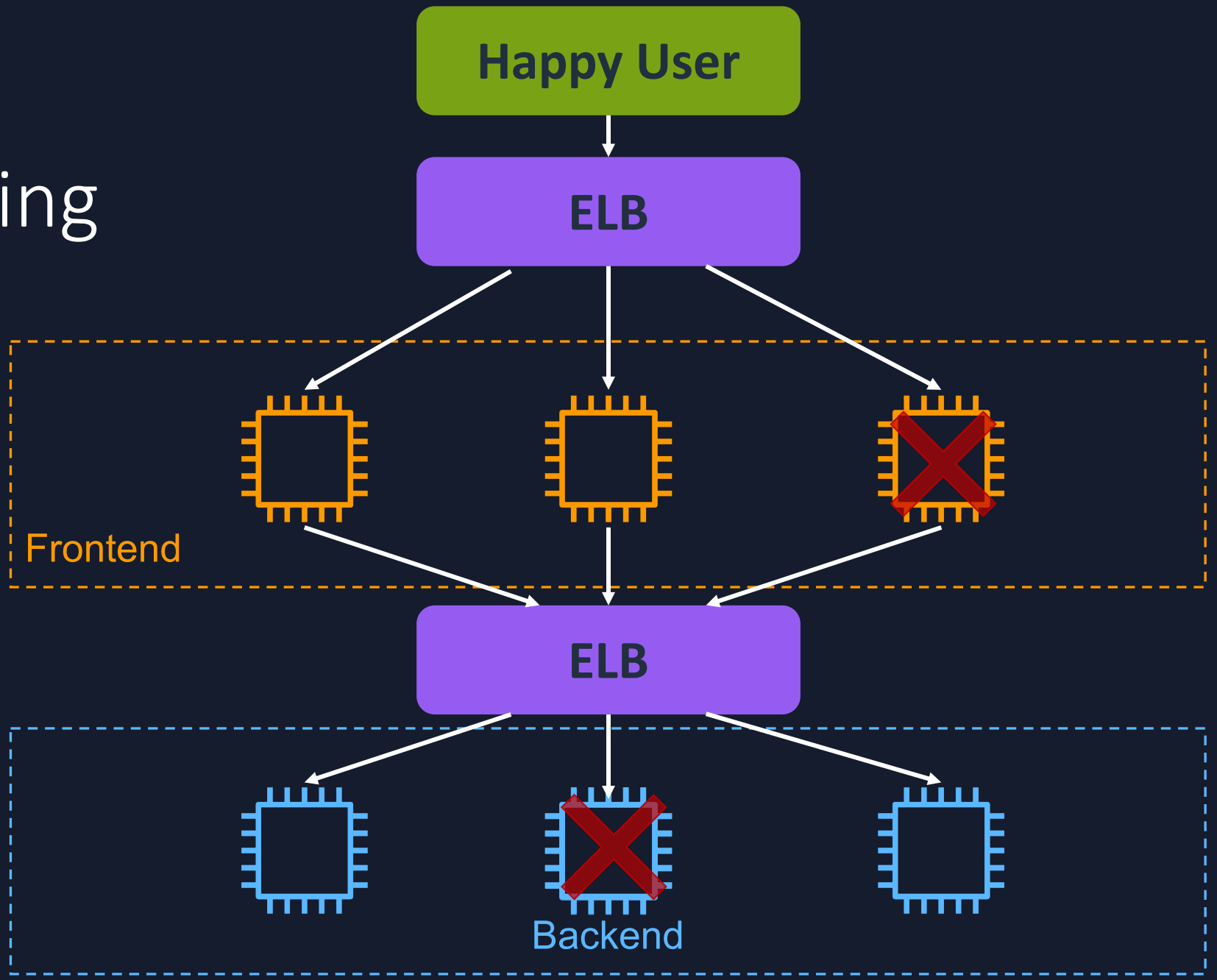
- Decoupling Workflows
- SQS
- Dead-letter Queues
- SQS FIFO
- SNS
- API Gateway
- AWS Batch
- Amazon MQ
- AWS Step Functions

Decoupling Workflows

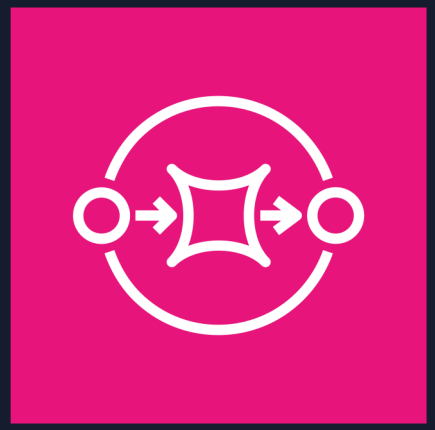
Tight Coupling



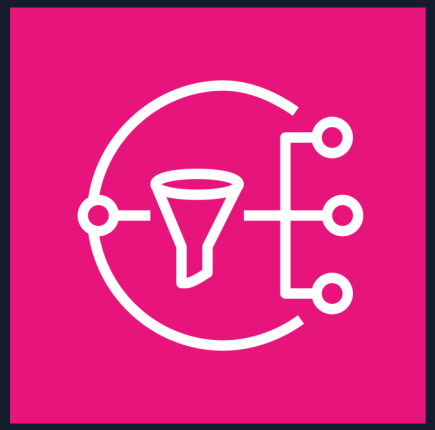
Loose Coupling



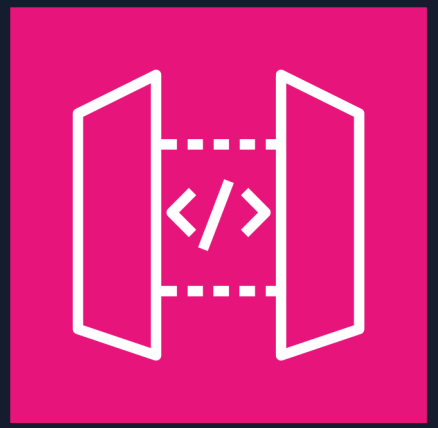
Upcoming Services



Amazon Simple Queue Service (Amazon SQS)

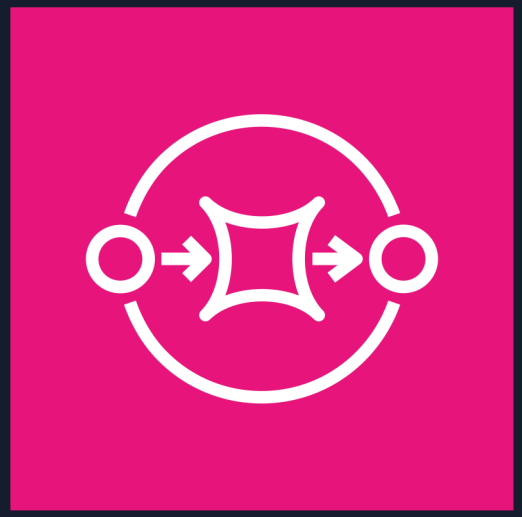


Amazon Simple Notification Service (Amazon SNS)



Amazon API Gateway

Simple Queue Service (SQS)



What is poll-based messaging?

What is SQS?

- Simple Queue Service (SQS) is a messaging queue that allows asynchronous processing of work
- One resource will write a message to an SQS queue, and then another resource will retrieve that message from SQS

SQS Settings

Delivery Delay

Default is 0; can be set up to 15 minutes.

Message Size

Messages can be up to 256 KB of text in any format.

Encryption

Messages are encrypted in transit by default, but you can add at-rest.

Message Retention

Default is 4 days; can be set between 1 minute and 14 days.

Long vs. Short

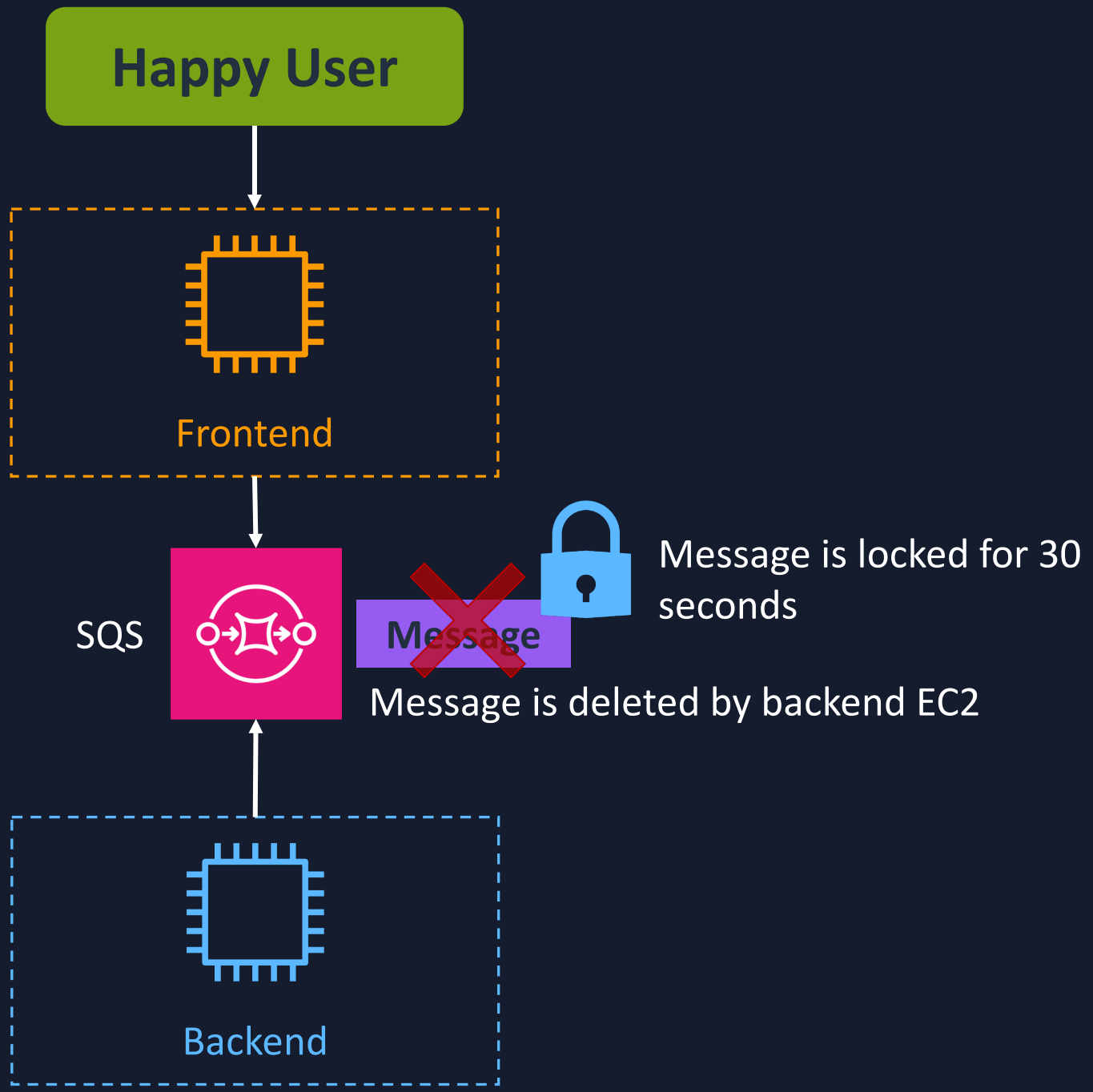
This can be a trigger for autoscaling.

Long polling isn't the default, but it should be.

Queue Depth

This can be a trigger for autoscaling.

Visibility Timeout



Dead-Letter Queues (DLQ)

Dead-Letter Queues (DLQ)

- A dead-letter queue (DLQ) is a special type of message queue that temporarily stores messages that a software system cannot process due to errors
- A dead-letter queue specifically stores erroneous messages that have no destination, or which can’t be processed by the intended receiver
- Technically they are just SQS queues

https://aws.amazon.com/what-is/dead-letter-queue/

Dead-Letter Queues Benefits

Alarm

Configure alarms based on message availability counts

Logs

Quickly identify logs to investigate for exceptions

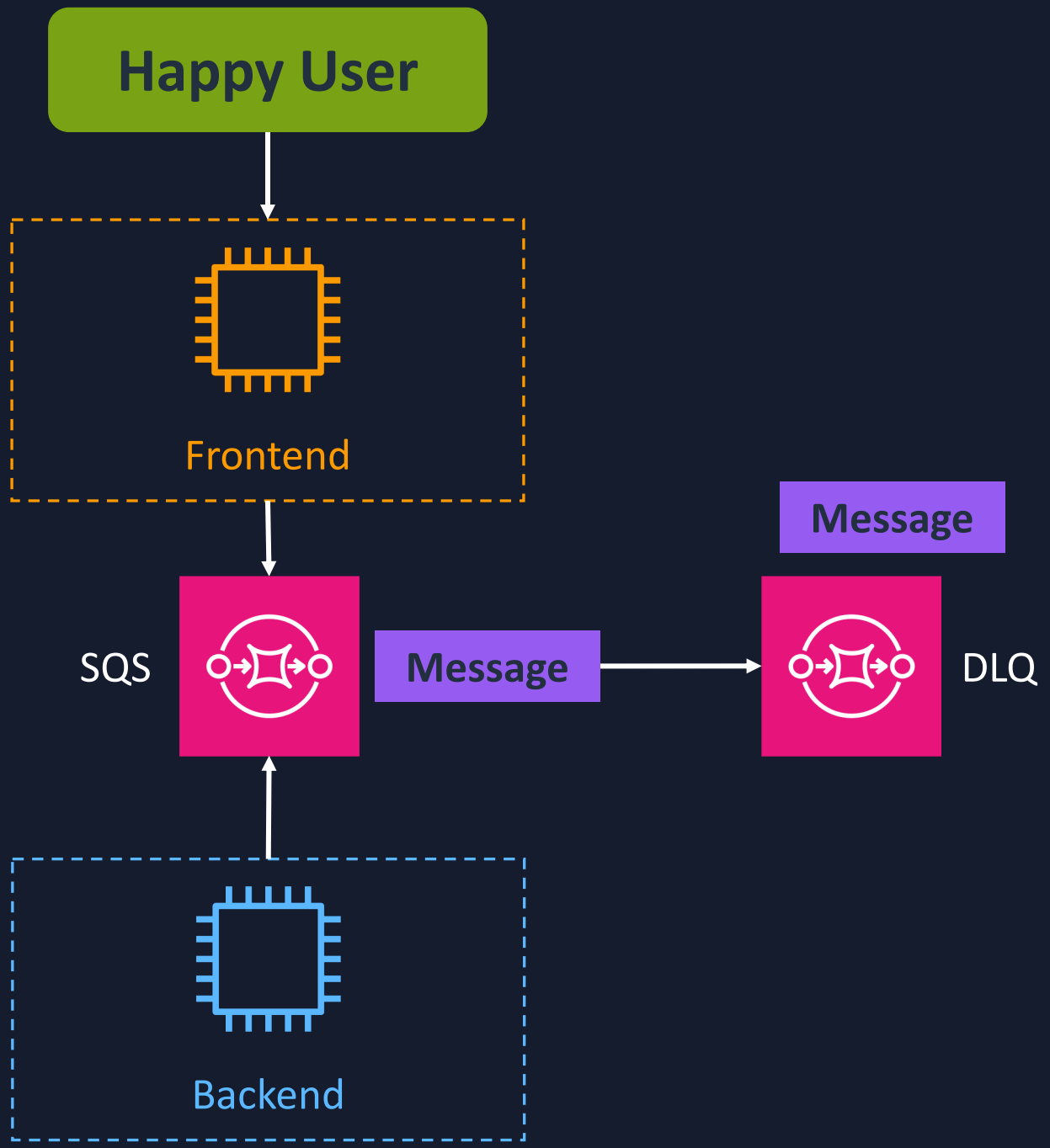
Errors

Analyze SQS message contents for errors

Permissions

Troubleshoot consumer permissions

Visibility Timeout



SQS FIFO

SQS Message Ordering

Standard

- Best-effort ordering
- Duplicate messages
- Nearly unlimited transactions per second

VS

FIFO

- Guaranteed Ordering
- No message duplication
- 300 transaction per second
- Batching can achieve up to 3,000 messages per second, per API call!

FIFO High Throughput

- Process up to 9,000 transaction per second, per API action without batching
- Up to 90,000 messages per second by using batching APIs

FIFO Queues

Performance

- FIFO queues do not have the same level of performance
- Can enable some features to make them more efficient; not going to perform the same in the end

Not the Only Way

- You can order messages with SQS standard, but it's on you to do it
- If you need your messages to be ordered, use FIFO queues

Message deduplication ID

- Token used for deduplication of sent messages
- Matching deduplication IDs are not delivered during a deduplication interval

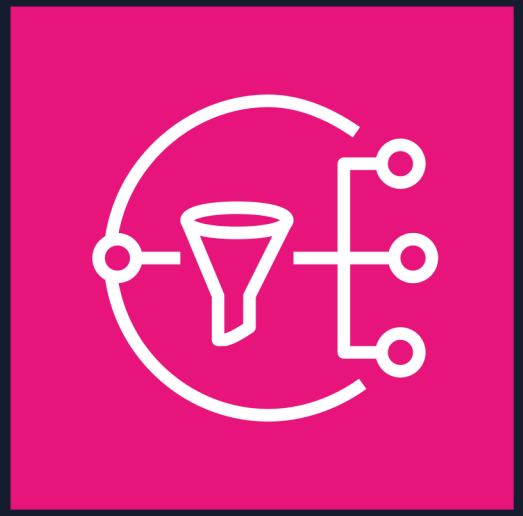
Message Group ID

- This ensures messages are processed one by one in a strict order based on the group

Cost

- It costs more since AWS must spend compute power to deduplicate messages

Simple Notification Service (SNS)



What is SNS?

- What is push-based messaging?

- Any message sent by a producer that arrives at the SNS topic will immediately be sent to all consumers

What is SNS?

- It is a pushed-based messaging service in AWS
- Allows you set up an application design where you can proactively deliver messages to the endpoints that are subscribed to it such as email, SQS queues, or HTTP clients that are subscribed to the topic
- Major use case: alert systems or people in real-time if a certain metric is crossed
- Big takeaway: one message can be sent to many receivers (one-to-many)

SNS Settings and Quotas

Subscribers

- Kinesis Data Firehose, SQS, Lambda, email, HTTP(S), SMS, platform application endpoint

Message Size

- Messages can be up to 256 KB of text in any format

DLQ Support

- Messages that fail to be delivered can be stored in an SQS DLQ

FIFO or Standard

- FIFO only supports SQS FIFO queues as a subscriber

Encryption

- Messages are encrypted in transit by default, but you can add at-rest via AWS KMS

Access Policy

- A resource policy can be added to a topic, like S3 bucket policies
- Useful for cross-account access

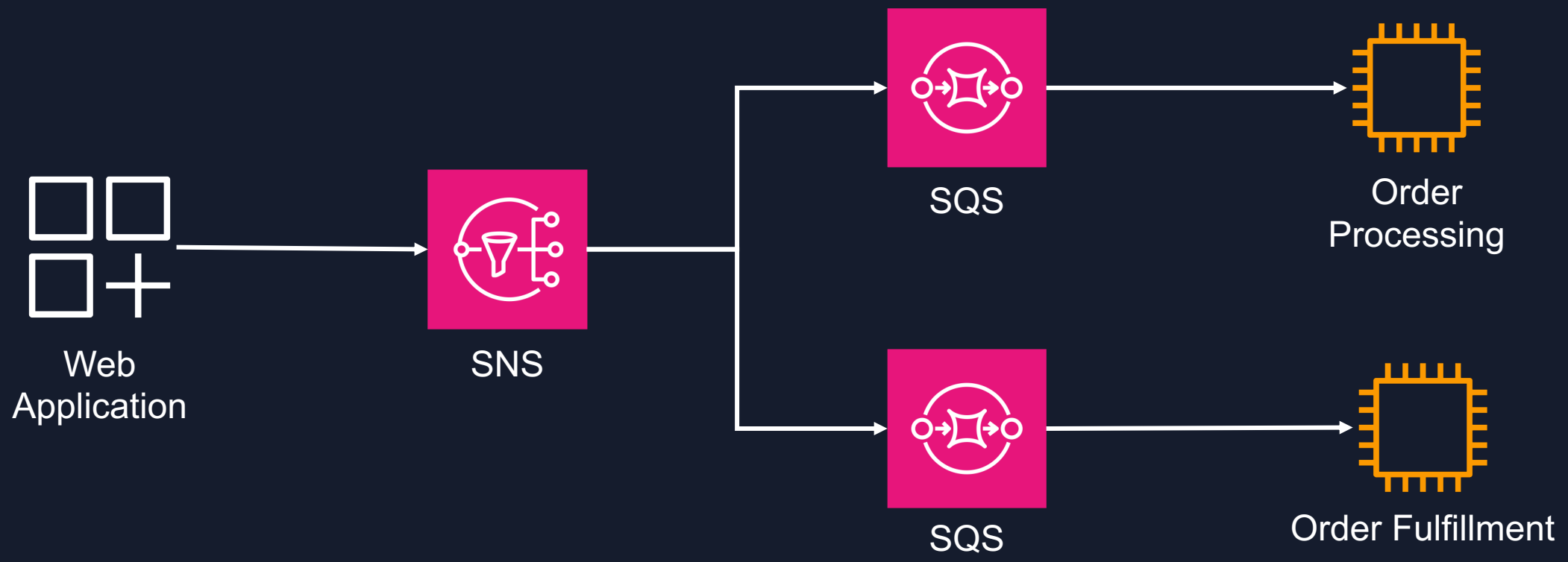
Large Message Payloads

- The SNS Extended Library allows for sending messages up to 2 GB in size
- The payload is stored in Amazon S3, then SNS pushed a reference to the object

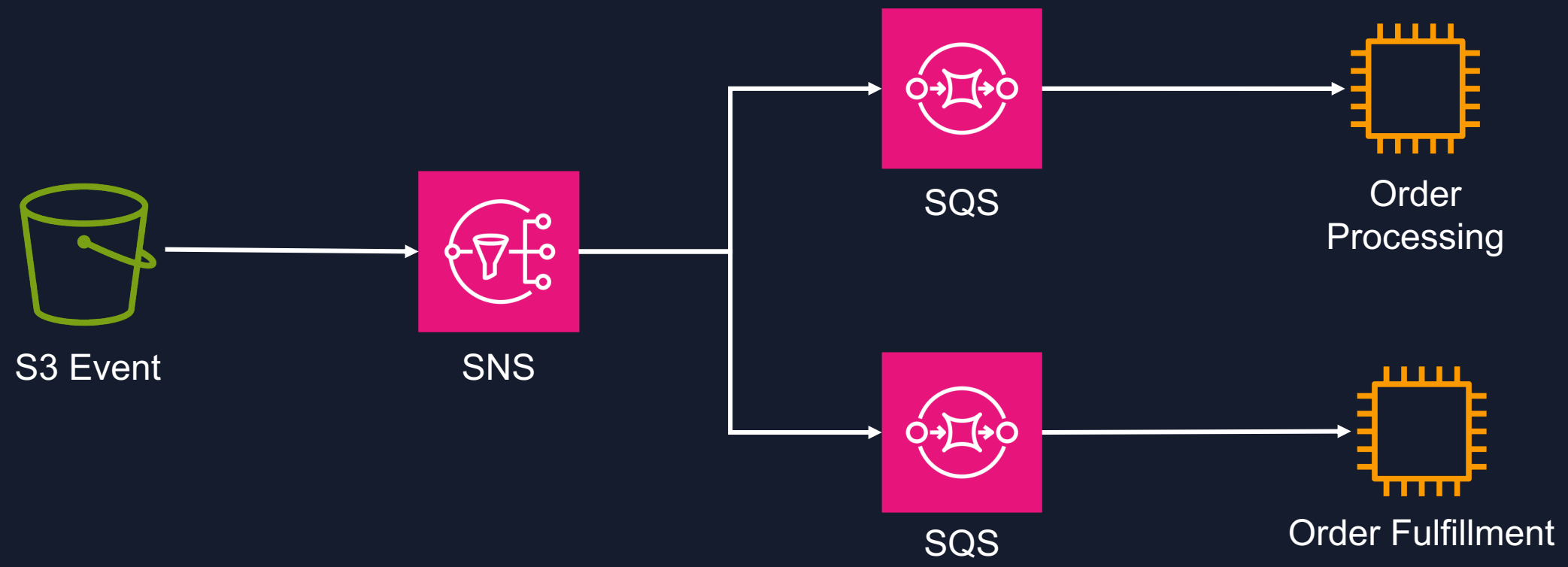
SNS Fanout

- Messages published to SNS topic are replicated to multiple endpoint subscriptions
- Allows for fully decouples parallel asynchronous processing
- Be familiar with the following architectures

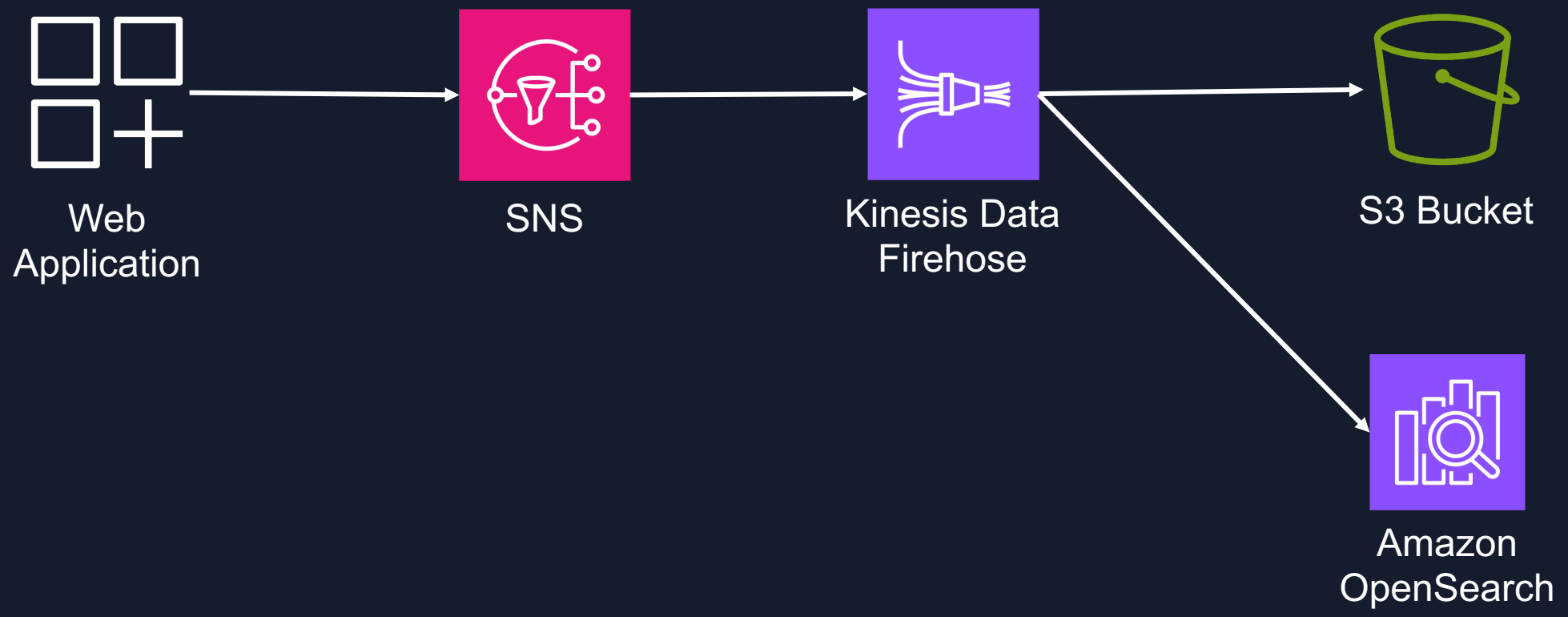
SNS Architectures – A Very Common Architecture



SNS Architectures – S3 Events



SNS Architectures – Analytics

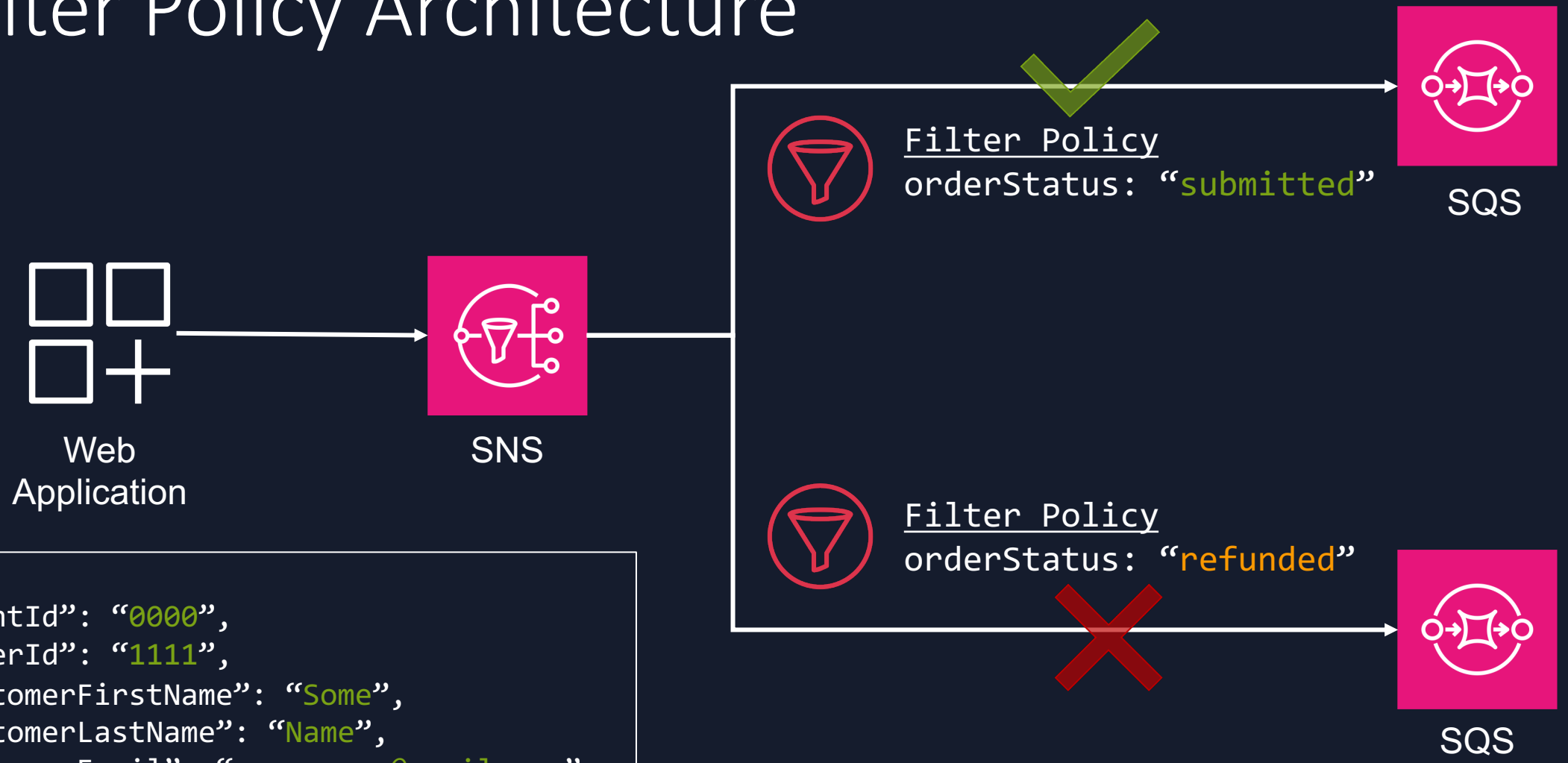


Message Filtering

- By default, every message published to a topic is sent to all subscribers

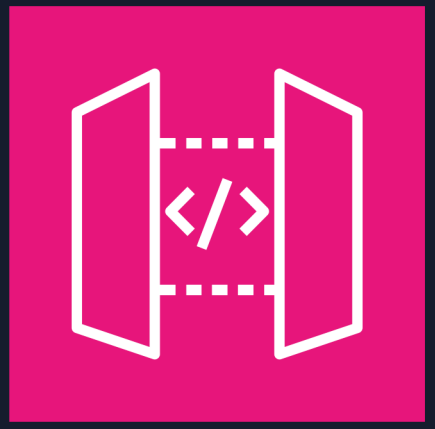
- Filter policies use JSON to define which messages get sent to specific subscribers

Filter Policy Architecture




```
{
    “eventId”: “0000”,
    “orderId”: “1111”,
    “customerFirstName”: “Some”,
    “customerLastName”: “Name”,
    “customerEmail”: “some.name@email.com”,
    “orderStatus”: “submitted”
}
```

Amazon API Gateway



What is Amazon API Gateway?

- Fully managed service that allows you to easily publish, create, maintain, monitor, and secure your API
- Integrates with many AWS services

API Gateway Features

Security

- Allows you to protect your endpoints by attaching a WAF

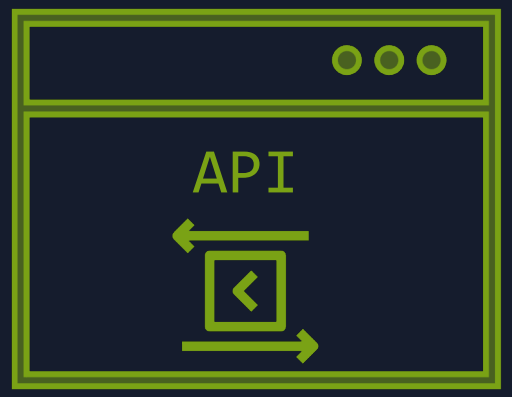
Stop Abuse

- Implement DDoS protection

Ease of Use

- Simple to get started with

API Options



REST API

API keys, per-client throttles, validation of requests, WAF integration

HTTP API

Simpler option than REST API, cheaper, minimal features

WebSocket API

collection of WebSocket routes integrated with Lambda functions, HTTP endpoints, and other AWS services

Endpoint Types

- **Edge-Optimized:** Default option. API requests get sent through a CloudFront edge location. Best for global users
- Regional: Perfect for clients that reside in the same specific region. Ability to also leverage CloudFront if required
- Private: Only accessible via VPCs using interface VPC endpoints

Securing Your APIs

- User authentication can be accomplished to control access to your APIs
- Authentication methods include IAM roles, Amazon Cognito, or even your own custom authorizer (Lambda functions)
- Edge-optimized endpoints require ACM certs in us-east-1
- Regional endpoints require ACM certs in the same region
- You can leverage AWS WAF in front of your API for DDoS protection

AWS Batch

What is AWS Batch?

Batched Workloads

- Allows you to run batch computing workloads within AWS (run on EC2 or ECS/Fargate)

Automatically Provision and Scale

- Capable of provisioning accurately sized compute resources based on number of jobs submitted, and optimizes the distribution of workloads

Makes Things Simpler

- Removes any heavy lifting for configuration and management of infrastructure required for computing

No Install Required!

- Allows you to skip installation and maintenance of batch computing software, so you can focus on obtaining and analyzing the results

Important Components

Jobs

- Units of work that are submitted to AWS Batch (e.g., shell scripts, executables, and Docker images)

Job Definitions

- Specify how your jobs are to be run (essentially, the blueprint for the resources in the job)

Job Queues

- Jobs get submitted to specific queues and reside there until scheduled to run in a compute environment

Compute Environment

- Set of managed or unmanaged compute resources used to run your jobs

Fargate or EC2 Environments

Fargate

- Recommended approach for MOST workloads
- Require fast start times (< 30 seconds)
- Require 16 vCPUs or less
- Require no GPUs
- Require 120 GiB of memory or less

VS

EC2

- Need more control over instance selection
- Require GPUs
- Require Elastic Fabric Adapter
- Require custom AMIs
- High levels of concurrency

Lambda or AWS Batch

Time Limits

- AWS Lambda currently has 1 15-minute execution time limit; Batch does not have this

Disk space

- Lambda has limited disk space, and EFS requires functions live within a VPC

Runtime Limitations

- Lambda is fully serverless, but it has natively limited runtimes; Big tradeoff

Batch Runtimes

- Batch uses Docker, so any runtime can be used

Managed and Unmanaged Compute Environments

Managed

- AWS Batch manages capacity and instance types
- Compute resource specs are defined when environment is created
- ECS instances are launched into VPC subnets
- Default is the most recent and approved Amazon ECS AMI
- You can use your own AMI
- Leverage Fargate, Fargate Spot, and regular Spot instances

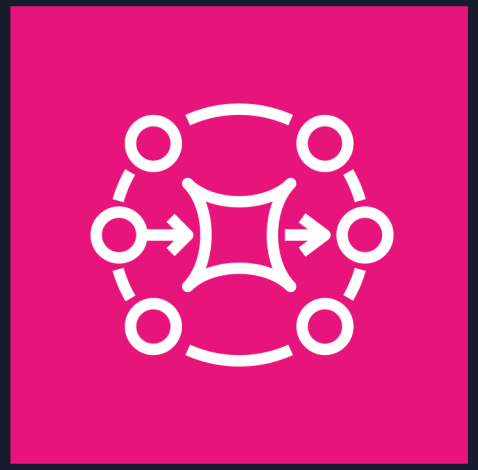
VS

Unmanaged

- You manage your own resources entirely
- AMI must meet Amazon ECS AMI specs
- You manage everything
- Less commonly used compared to managed
- Good choice for extremely complex or specific requirements

Amazon MQ

Brokering Messages with Amazon MQ



Amazon MQ Overview

- Message broker service allowing easier migration of existing applications to the AWS Cloud
- Leverage multiple programming languages, operating systems, and messaging protocols
- Currently supports both Apache ActiveMQ or RabbitMQ engine types
- Allows you to easily leverage existing apps without managing and maintaining your own system

SNS with SQS or Amazon MQ

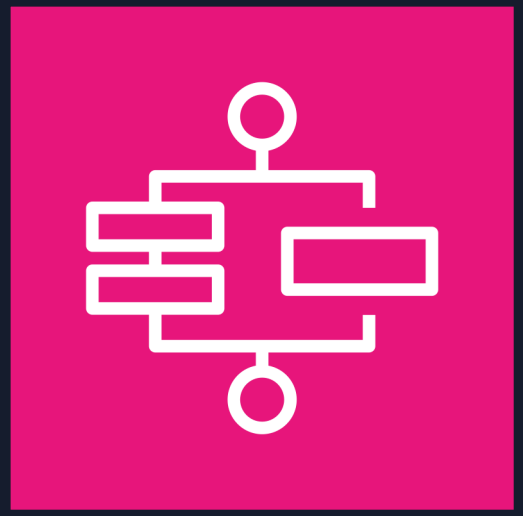
- Each offers architectures with topics and queues. Allows for one-to-one or one-to-many messaging designs
- If migrating existing applications with messaging systems in place, you might consider Amazon MQ
- If creating new applications, look at SNS and SQS. These are simpler to use and are highly scalable
- Amazon MQ requires private networking like VPC, Direct Connect, or VPN
- SNS and SQS are publicly accessible by default

Amazon MQ Brokers

Amazon MQ	Amazon MQ for ActiveMQ	Amazon MQ for RabbitMQ
Amazon MQ offers highly available architectures to minimize downtime during maintenance Architecture depends on the broker engine type	With active/standby deployments, one instance will remain available at all time Configure network or brokers with separate maintenance windows	Cluster deployments are logical grouping of three broker nodes across multiple AZs sitting behind a Network Load Balancer

Amazon MQ

AWS Step Functions



What is AWS Step Functions?

Orchestration

- Serverless orchestration service combining different AWS services for business applications

Graphical Console

- Comes with a graphical console for easier application workflow views and flows

Component

- State machines and tasks

State Machine

- A particular workflow with different event-driven steps

Task

- Specific states within a workflow (state machine) representing a single unti of work

States

- Every single step within a workflow is considered a state

Two Different Types of Workflows

Standard

- Have an exactly-once execution
- Can run for up to one year
- Useful for long-running workflows that need to have an auditable history
- Rates up to 2,000 execution per second
- Pricing based per state transition

VS

Express

- At least-once workflow execution
- Can run for up to five minutes
- Useful for high-event-rate workloads
- Example use is IoT data streaming and ingestion
- Pricing based on number of executions, durations, and memory consumed

States and State Machines

Flexible

- Leverage states to either make decisions based on input, perform certain actions, or pass output

Language

- States and workflows are defined in Amazon States Language

States

- Elements within your state machines. They are referred to by a name

Example

- Think about an online pickup order: Each step in that workflow is considered a state

Different States

- **Pass:** Passes any input directly to its output — no work done
- **Task:** Single unit of work performed (e.g., Lambda, Batch, and SNS)
- **Choice:** Adds branching logic to state machines
- **Wait:** Creates a specified time delay within the state machine
- **Succeed:** Stops executions successfully
- **Fail:** Stops executions and marks them as failures
- **Parallel:** Runs parallel branches of executions within state machines
- **Map:** Runs a set of steps based on elements of an input array

Checkpoint

If the visibility timeout on a message in an SQS queue expires, what happens to the message?

1. An SNS Topic is notified.
2. It's now available in the queue to be retrieved.
3. It's emailed to sqs-help@amazon.com.
4. It's deleted.

API Gateway is commonly built __ your applications.

1. in front of
2. within
3. using
4. behind

You need to migrate an existing RabbitMQ-based application from on-premises to AWS with the least impact to functionality and design. Which service would be the best choice?

1. AWS Application Messaging Service
2. Amazon Managed Broker
3. Amazon MQ
4. AWS Batch

Which layers of our applications need to be loosely coupled?

1. Internal and external
2. Just internal
3. None — it's bad practice to loosely couple applications
4. Just external

If your application needs to process 5,000 messages per second, which type of SQS queue would you use?

1. SQS FIFO
2. SQS paired with an EC2 Auto Scaling Group
3. SQS Performance
4. SQS Standard

Which AWS service would you choose for serverless orchestration of long-running (up to 1 year) workflows that can integrate with several other AWS services?

1. Amazon MQ
2. AWS EC2 Spot Instances
3. AWS Step Functions
4. AWS Lambda

SNS is a _____-based messaging service.

1. Poll
2. Push
3. Grab
4. Pull

Which tool can be used to monitor the queue depth of a DLQ?

1. S3
2. SQS
3. CloudWatch
4. CloudTrail

You need an AWS managed service to execute long-running, event-driven, batched workloads that are expected to take 30 minutes to complete. Which service best meets your needs?

1. AWS Lambda
2. AWS Batch
3. AWS AppSync
4. API Gateway

What type of service is SQS?

1. Machine learning tool
2. Compute resource
3. Database
4. Messaging queue

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